

Marketing Analytics

Chapter 5 · Learning from Observational Data

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Part 1 · Observational data I

Before we start

Paper quiz on Chapter 4 (*10 min*)

- Individual, closed book
- Counts toward participation

Assignment 2 is running

- Deadline: Group A Sun, Nov 8 · Group B Wed, Nov 11
- Debrief at the start of next session

Today's session

Observational data I (*today*)

- 5.1 Smart Mix without randomization
- 5.2 From correlation to a line
- 5.3 Reading a regression
- 5.4 Adjusting for confounders
- 5.5 Categorical predictors
- 5.6 Prediction vs. explanation

Observational data II (*next session*)

- 5.7 Functional form and elasticities
- 5.8 Logistic regression
- 5.9 Diagnostics that change decisions
- 5.10 Difference-in-differences

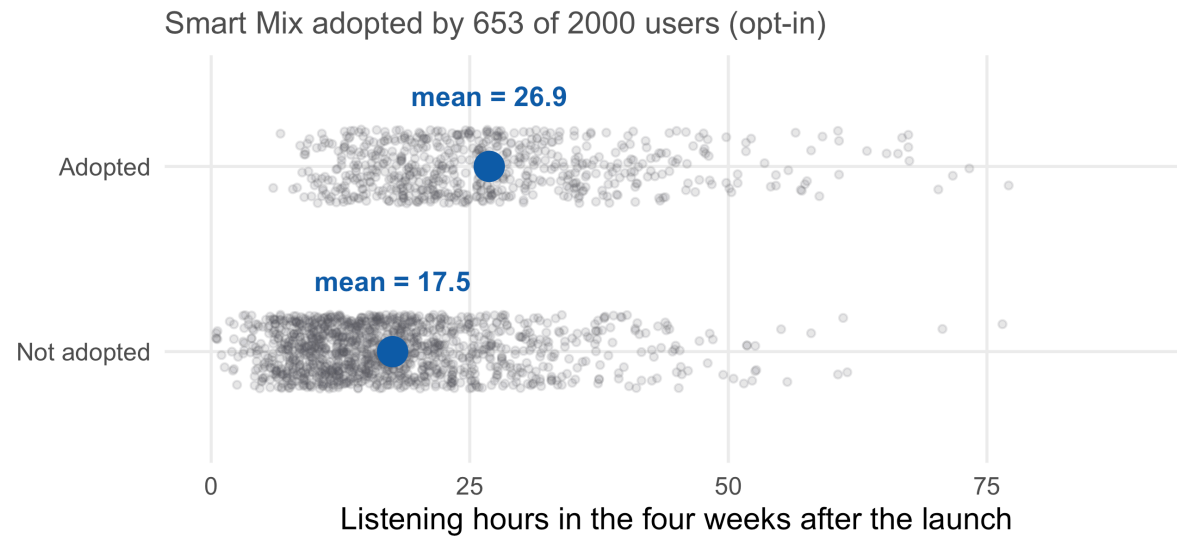
Driving question: *What if we cannot randomize?*

5.1 Smart Mix without randomization

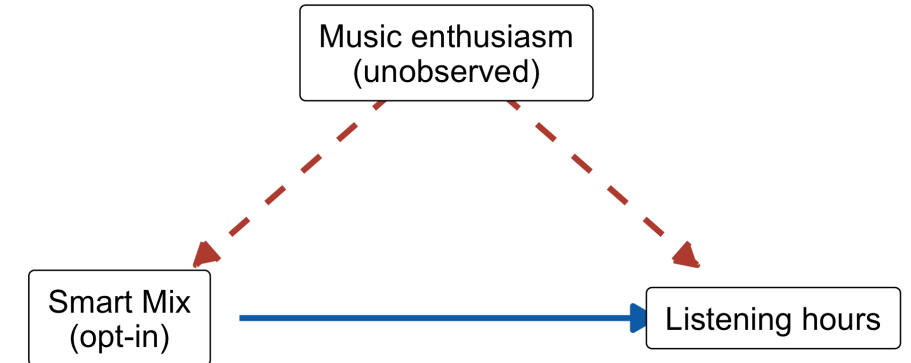
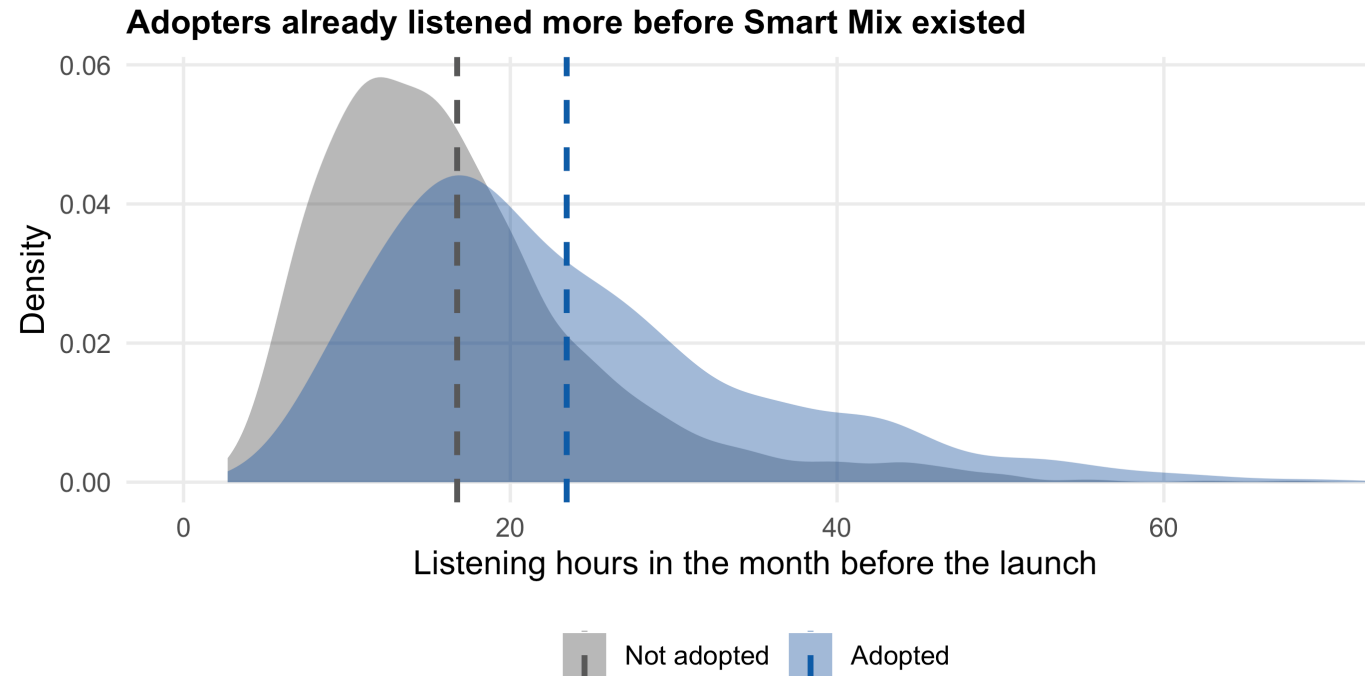
A different history

- Chapter 4: A/B test, 400 users → **+3.2 hours**
- Now: no experiment. Smart Mix launched as an **opt-in** feature
- Four weeks later: data of **2,000 users**, 653 adopters

The dashboard says: adopters listen **9.3 hours more** — three times the experimental effect. Would you report this number to management?



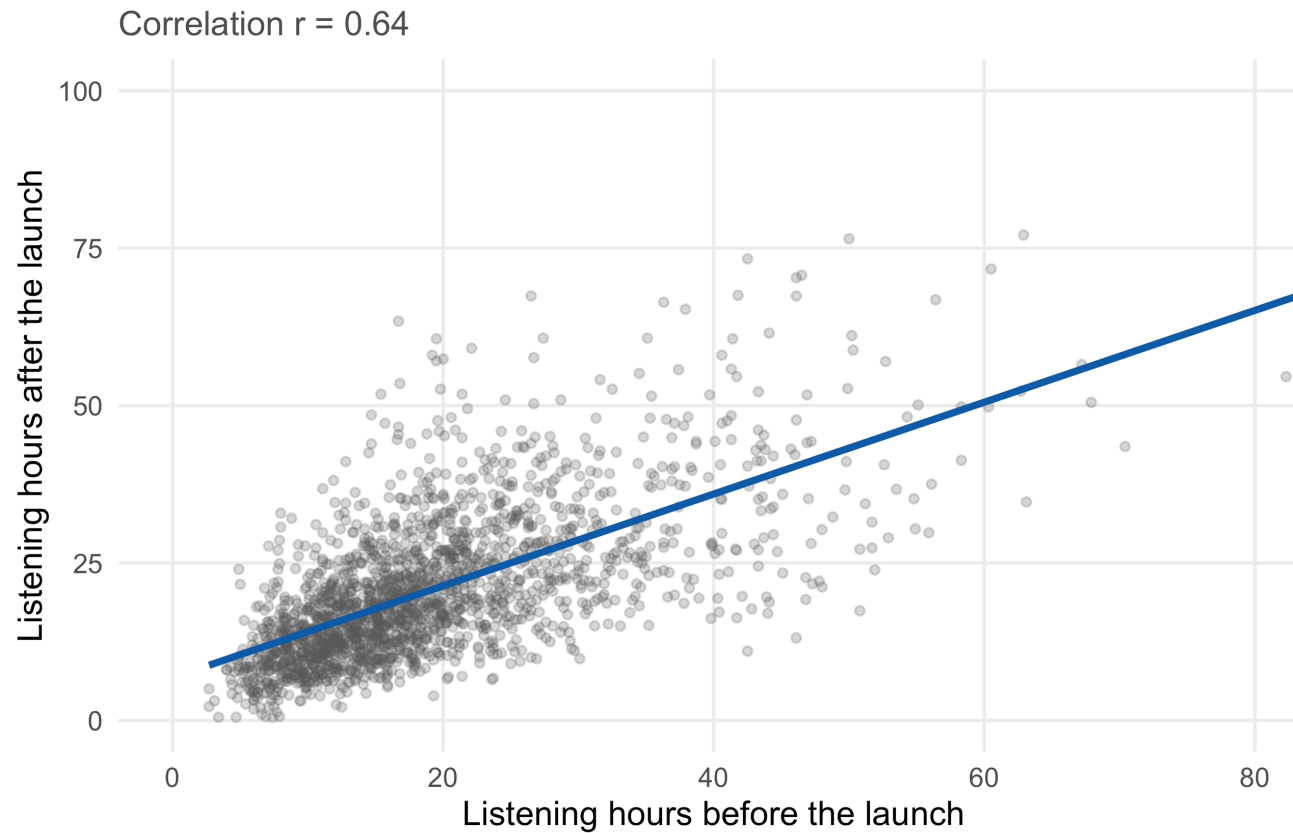
Adopters were different before Smart Mix existed



Same costume, new villain: **music enthusiasm** drives adoption *and* listening. Without a coin flip, we have one tool left: **adjustment**.

5.2 From correlation to a line

Two variables, one line



$$\text{hours} = \beta_0 + \beta_1 \cdot \text{prior hours} + \varepsilon$$

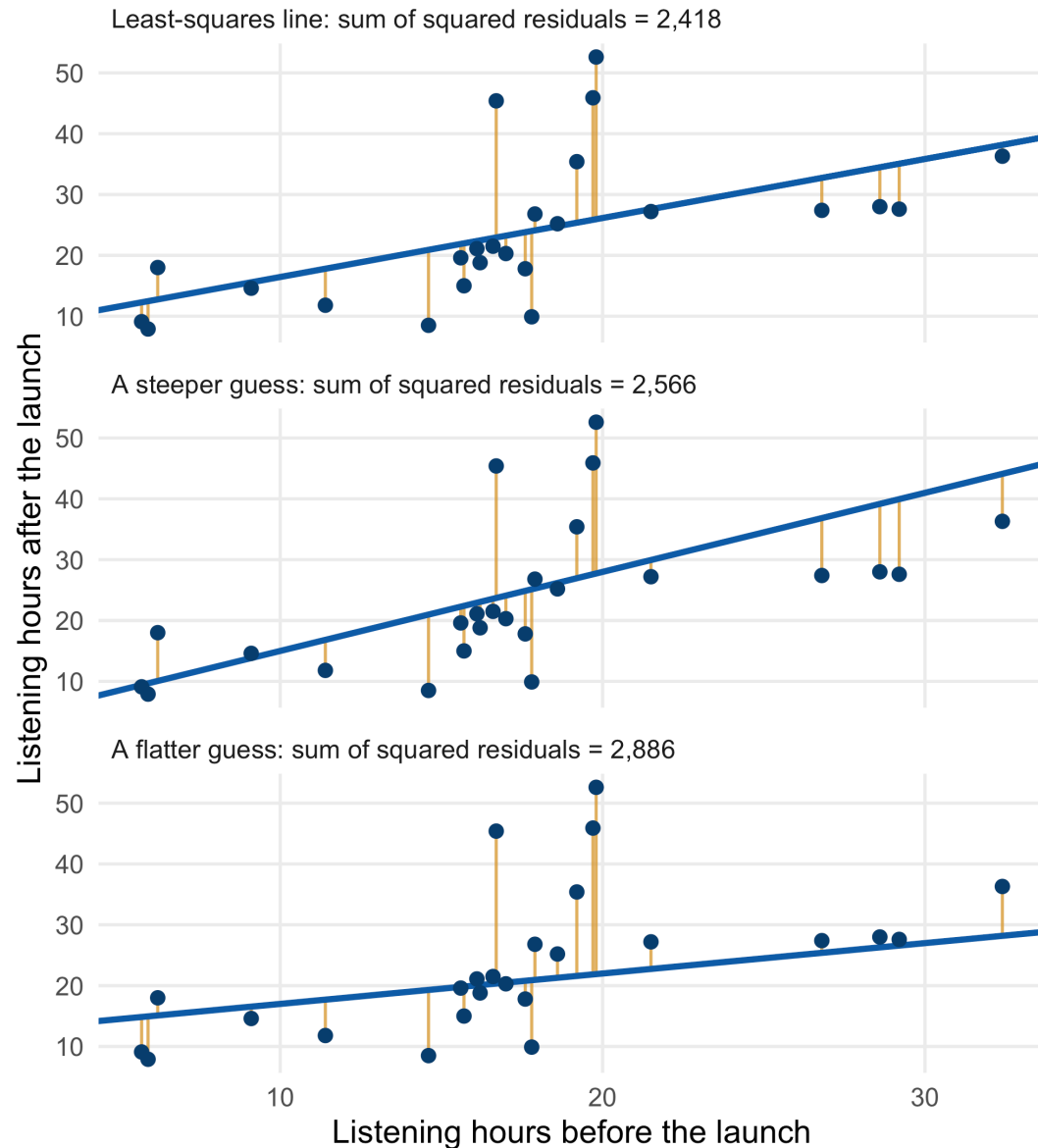
- β_0 intercept · β_1 slope · ε residual
- Correlation: *that* they move together · regression: *how much*

```
1 coef(lm(hours ~ prior_hours, data = smartmix_obs))
```

```
(Intercept) prior_hours
        6.7849         0.7291
```

Each extra hour before the launch **goes with** 0.73 hours after it — *goes with*, not *causes*.

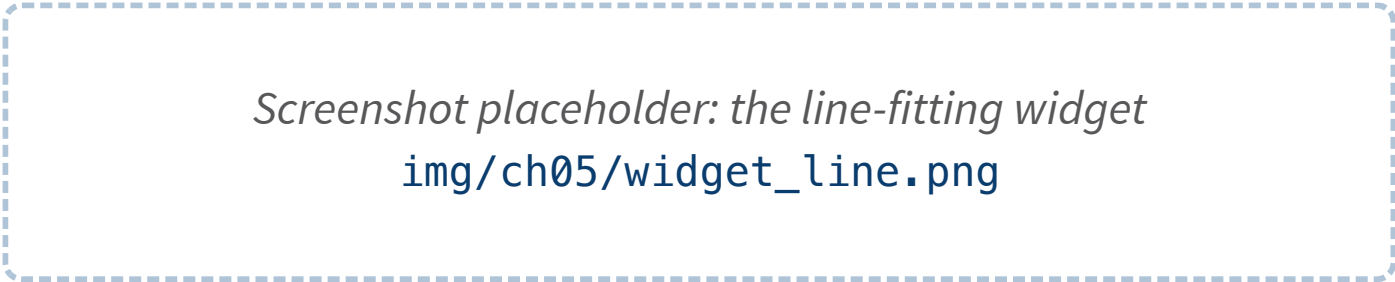
Which line? Least squares



- Residual = observed – predicted (orange segments)
- The regression line **minimizes the sum of squared residuals**
- Squaring: big misses count a lot; above and below the line count alike
- `lm()` = “linear model”: does exactly this

Live demo: fit the line yourself — course book, §5.2

<https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-line>

Screenshot placeholder: the line-fitting widget

[img/ch05/widget_line.png](#)

5.3 Reading a regression

The full output

```
1 model_naive <- lm(hours ~ smart_mix, data = smartmix_obs)
2 summary(model_naive)
```

Call:

```
lm(formula = hours ~ smart_mix, data = smartmix_obs)
```

Residuals:

Min	1Q	Median	3Q	Max
-20.88	-7.64	-2.06	4.76	101.22

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.541	0.300	58.5	<2e-16 ***
smart_mix	9.341	0.525	17.8	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11 on 1998 degrees of freedom

Multiple R-squared: 0.137, Adjusted R-squared: 0.136

F-statistic: 316 on 1 and 1998 DF, p-value: <2e-16

Line by line

Output	Meaning
Estimate	Intercept = mean of non-adopters (18.4); <code>smart_mix</code> = difference adopters – non-adopters (9.3)
Std. Error	Uncertainty of each coefficient (Chapter 3)
t value, Pr(> t)	<i>t</i> -test of each coefficient against zero (Chapter 4): signal / noise
Residual standard error	Typical distance of a user from the line, in hours
Multiple R-squared	Share of the variation in hours explained by the model (14%)
F-statistic	The whole model against “no predictors”

With one 0/1 predictor, the regression is the comparison of two means — and its *t*-test is the two-sample *t*-test. `confint(model_naive)` gives the confidence intervals.

5.4 Adjusting for confounders

Holding prior listening constant

```
1 model_prior <- lm(hours ~ smart_mix + prior_hours, data = smartmix_obs)
2 coef(summary(model_prior))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.4508	0.41213	15.65	3.538e-52
smart_mix	4.9011	0.44252	11.08	1.039e-27
prior_hours	0.6623	0.01996	33.17	1.335e-192

The Smart Mix coefficient drops from **9.3** to **4.9**: now the difference between adopters and non-adopters **with the same prior listening**. Part of the naive gap belonged to prior listening — and is now attributed to it.

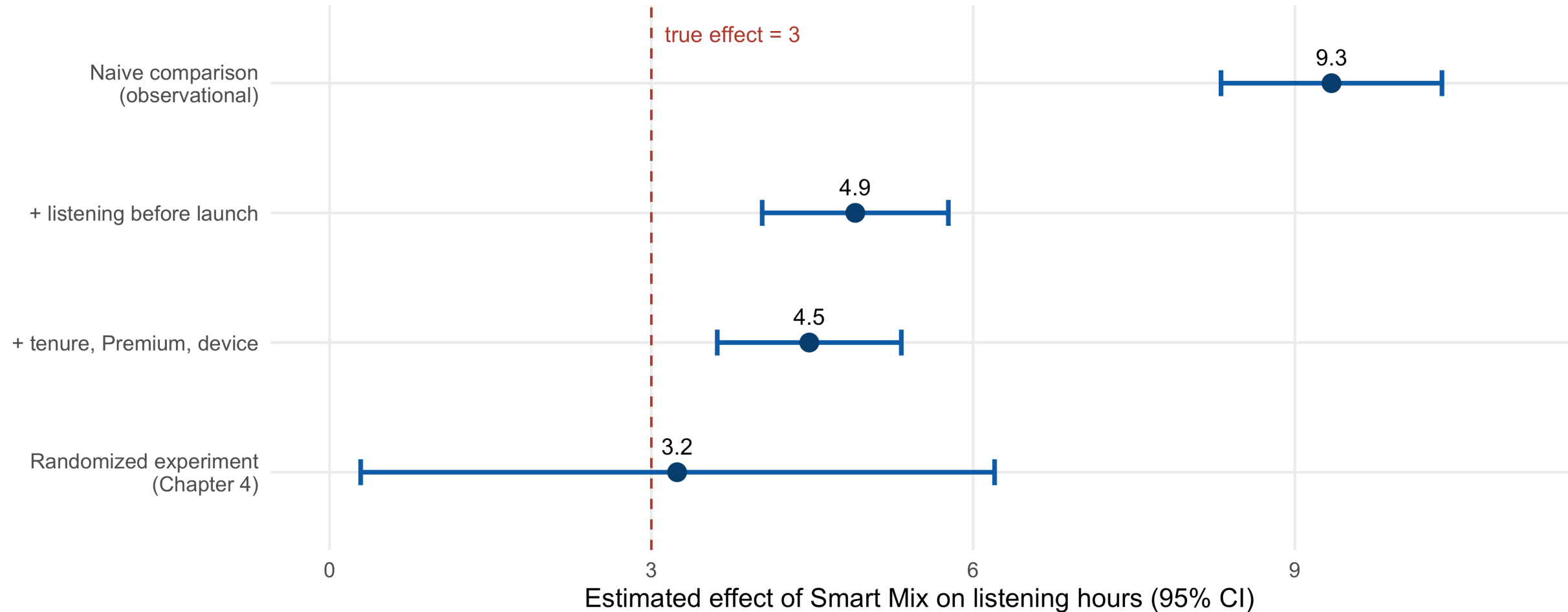
More variables, same logic

```
1 model_full <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device, data = smartmix_obs)
2 coef(summary(model_full))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.25472	0.54036	11.575	4.918e-30
smart_mix	4.47240	0.43747	10.223	6.013e-24
prior_hours	0.61258	0.02063	29.688	1.074e-160
tenure	0.03771	0.01128	3.344	8.397e-04
premium	2.92426	0.42572	6.869	8.610e-12
deviceAndroid	-1.47497	0.42230	-3.493	4.886e-04
deviceDesktop	-2.48802	0.57594	-4.320	1.638e-05

Each coefficient: the association with the outcome **holding all other variables in the model constant**. Tenure, Premium and device change little: **≈ 4.5 hours**.

Naive, adjusted – and the truth



Adjustment removed ~75% of the bias — **not all of it**. We adjusted for a *proxy* of enthusiasm, not for enthusiasm. Regression removes exactly the confounding your variables capture.

Live demo: how good must a proxy be?

Course book, §5.4 — set the **quality of the proxy** and the **strength of confounding**.

<https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-adjust>

Screenshot placeholder: the proxy widget
`img/ch05/widget_proxy.png`

With a 50% proxy, how much of the bias is left? And what happens at 0%?

Two warnings

Adjusting for the wrong variable makes it worse. A **mediator** (on the causal path) absorbs part of the effect you want. Only **confounders** — determined *before* the treatment, affecting treatment *and* outcome — belong in the model.

“Enough” is **untestable**. The assumption of no unobserved confounders cannot be checked in the data. Report *what* you adjusted for — and what you plausibly could not.

Your turn (5 min)

Add **age** to the full model. Does the Smart Mix coefficient change? Why (not)?

```
1 model_age <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device + age,  
2                 data = smartmix_obs)  
3 coef(summary(model_age))
```

Then: which variable in the data set should you **never** adjust for when estimating the effect of Smart Mix on listening — and why?

5.5 Categorical predictors

Dummies and the reference category

```
1 coef(model_full)[c("premium", "deviceAndroid", "deviceDesktop")]
```

```
premium deviceAndroid deviceDesktop  
2.924      -1.475      -2.488
```

- **0/1 variable** (`premium`): coefficient = difference between the two groups, other variables held constant
- **Several categories** (`device`): R picks a **reference** (first level: iOS) and creates one **dummy** per other category
- Each dummy = difference to the **reference**

Android users: **1.5 hours less** than iOS users; Desktop: **2.5 hours less** — holding everything else constant. *p*-values compare each category with the reference; “does device matter at all?” → `anova(model_without, model_full)` (the ANOVA of Chapter 4).

5.6 Prediction vs. explanation

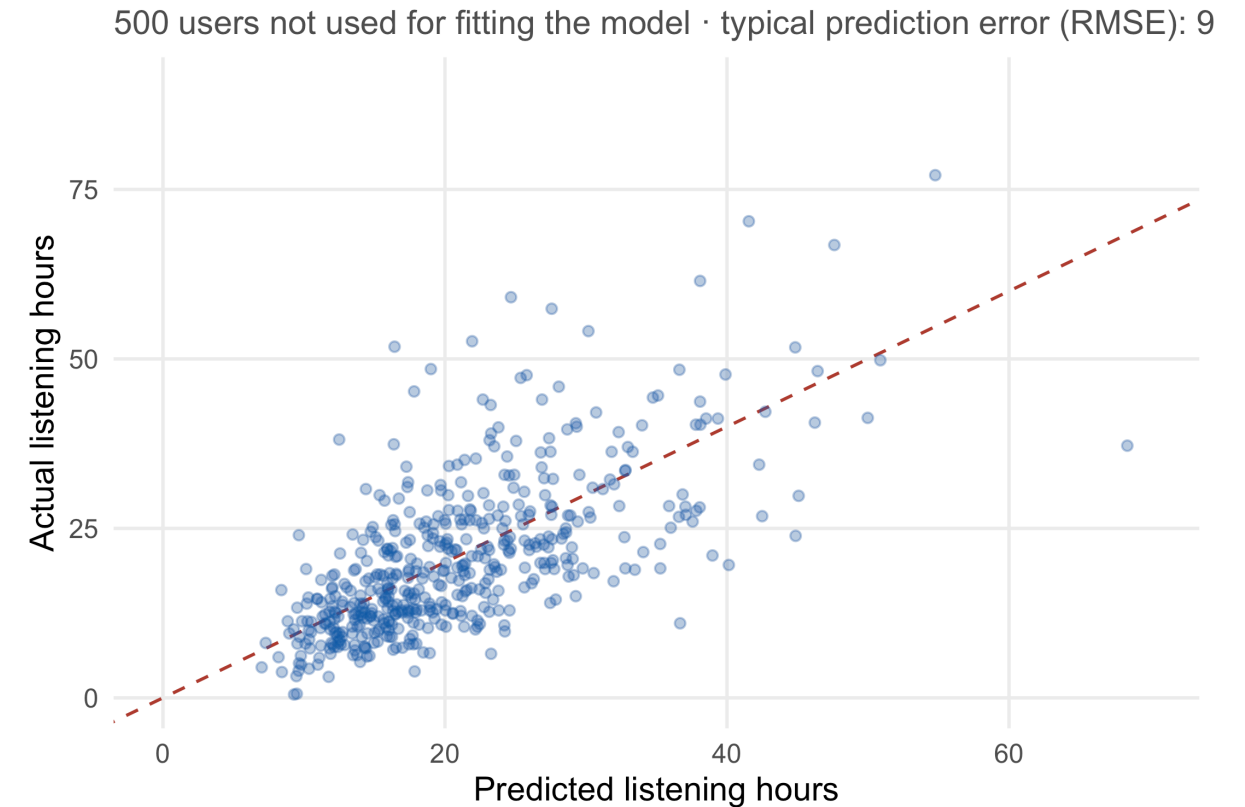
Predicting listening time for new users

```

1 set.seed(42)
2 train_ids <- sample(nrow(smarmix_obs), 1500)
3 train <- smarmix_obs[train_ids, ]
4 test <- smarmix_obs[-train_ids, ]
5 model_pred <- lm(hours ~ prior_hours + tenure + premium +
6                 device + smart_mix, data = train)
7 test$predicted <- predict(model_pred, newdata = test)
8 sqrt(mean((test$hours - test$predicted)^2)) # RMSE

```

[1] 9.425



For **prediction**, it does not matter *why* a variable is associated with the outcome — only how well the predictions hold up on **data the model has not seen**.

Two questions, two rules

Explanation (causal question): include confounders, exclude mediators, state what is unobserved. The **coefficient** matters; R^2 is secondary.

Prediction: include whatever predicts; judge by **out-of-sample errors**. Coefficients are not interpreted; the **predictions** matter.

A model can be a **good predictor and a bad explainer** at the same time. Where in your company would each mistake — reading causal effects off a prediction model, or judging a causal model by its R^2 — cost money?

Learning check (Part I)

1. Slope of hours on prior hours = 0.73. Interpret — and why is it not causal?
2. Adding a variable can increase, decrease, or leave the Smart Mix coefficient unchanged. When does each happen?
3. `planFamily` = 4.2 with reference “Free”. Interpret. How do you test whether the plan matters at all?
4. $R^2 = 0.71$ for predicting next month’s spending with 25 predictors on 300 customers. What do you ask?

Next session

Observational data II — beyond the straight line, and a quasi-experiment

- Functional form: when a line is wrong — **elasticities**
- Yes/no outcomes: logistic regression
- The three diagnostics that change decisions
- **Difference-in-differences** with real playlist data

Before next session: work through §5.1–5.6 on the course site. **Debrief of Assignment 2** at the start of the session.

Part 2 · Observational data II

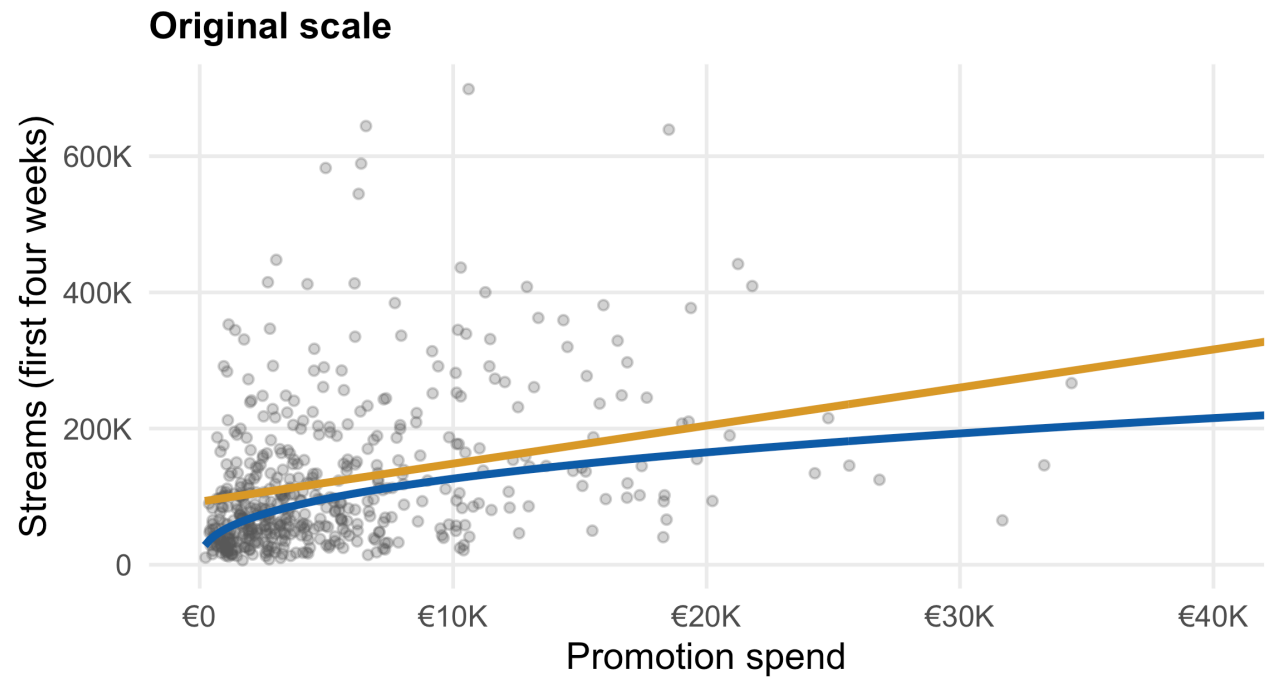
Before we start

Debrief Assignment 2 — analyzing experiments (*10 min*) **Paper quiz on Chapter 5, Part I** (§5.1–5.6, *8 min*)

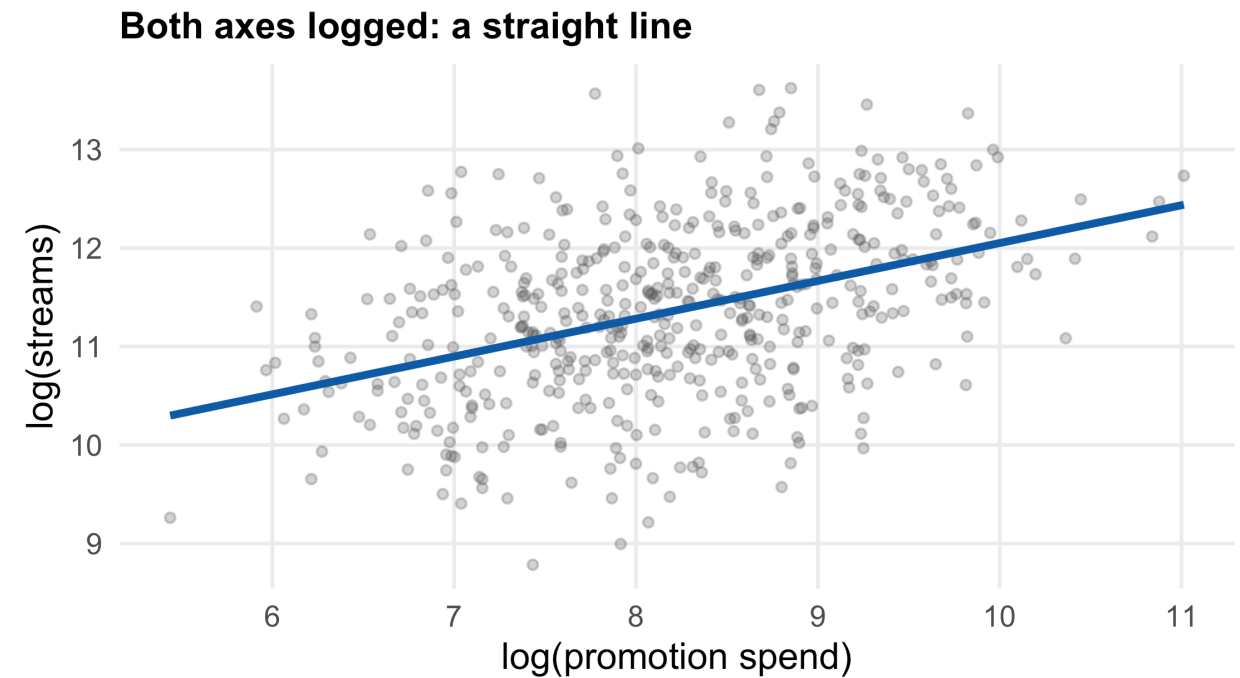
- Three short paper questions about *your* results
- Closed book, no laptop
- Individual, closed book
- Counts toward participation

5.7 Functional form: when the line is wrong

Diminishing returns



- Linear model: $\text{streams} = a + b \times \text{spend}$
- Log-log model: $\log(\text{streams}) = a + b \times \log(\text{spend})$



500 promotion campaigns: the first euros buy many streams, later euros fewer. A straight line cannot bend — on **log scales**, the cloud becomes a line again.

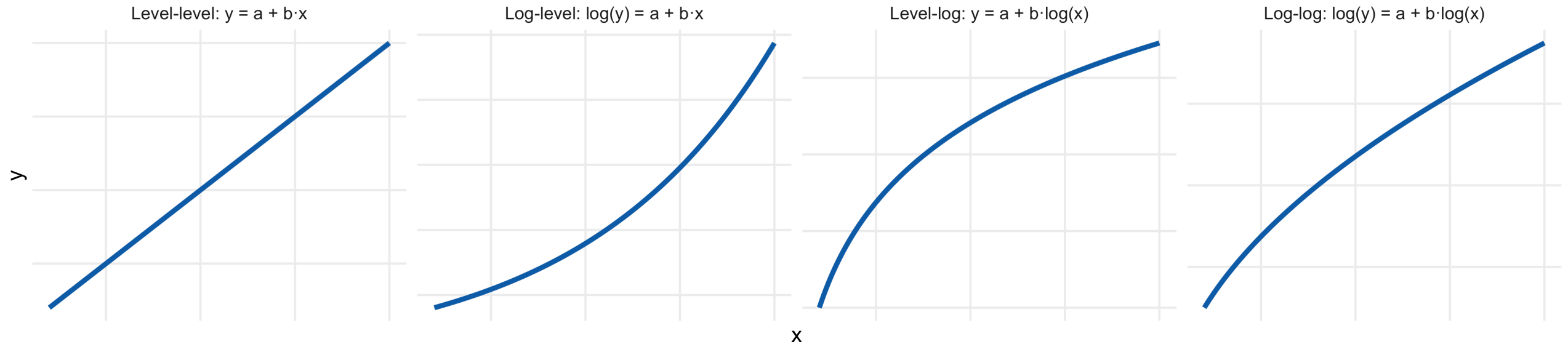
The log-log model: elasticities

```
1 model_loglog <- lm(log(streams) ~ log(spend) + log(followers) + placements + genre, data = campaigns)
2 coef(summary(model_loglog))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.8815	0.23461	12.282	1.946e-30
log(spend)	0.3310	0.02069	16.001	1.017e-46
log(followers)	0.5071	0.01571	32.279	1.137e-123
placements	0.1203	0.01195	10.067	8.320e-22
genreHipHop/Rap	0.1994	0.04761	4.188	3.336e-05
genreRock	-0.2555	0.05985	-4.270	2.349e-05
genreElectro/Dance	0.1026	0.05939	1.728	8.464e-02

$\log(\text{spend}) = 0.33$: a 1% increase in spend goes with 0.33% more streams; +10% spend $\rightarrow$ +3.3% **streams** — the same *percentage* at every budget. Elasticities have no units and are **comparable**: followers (0.51) matter more than the budget.

Four ways to combine levels and logs



Model	Equation	Reading β_1
Level-level	$y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \beta_1$ units of y
Log-level	$\log y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \approx 100\beta_1$ % of y (exact: $100(e^{\beta_1} - 1)\%$)
Level-log	$y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1/100$ units of y
Log-log	$\log y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1$ % of y (elasticity)

Your turn (5 min)

From the log-log model:

1. $\text{genreHipHop/Rap} = 0.20$. By what percentage do HipHop/Rap campaigns differ from Pop campaigns (the reference) — approximately and exactly?
2. A marketing manager proposes to double the promotion budget of every campaign. By roughly what percentage would streams increase, according to the model? Is this a causal claim?

Live demo: linear or log-log?

Course book, §5.7 — move the **true elasticity**.

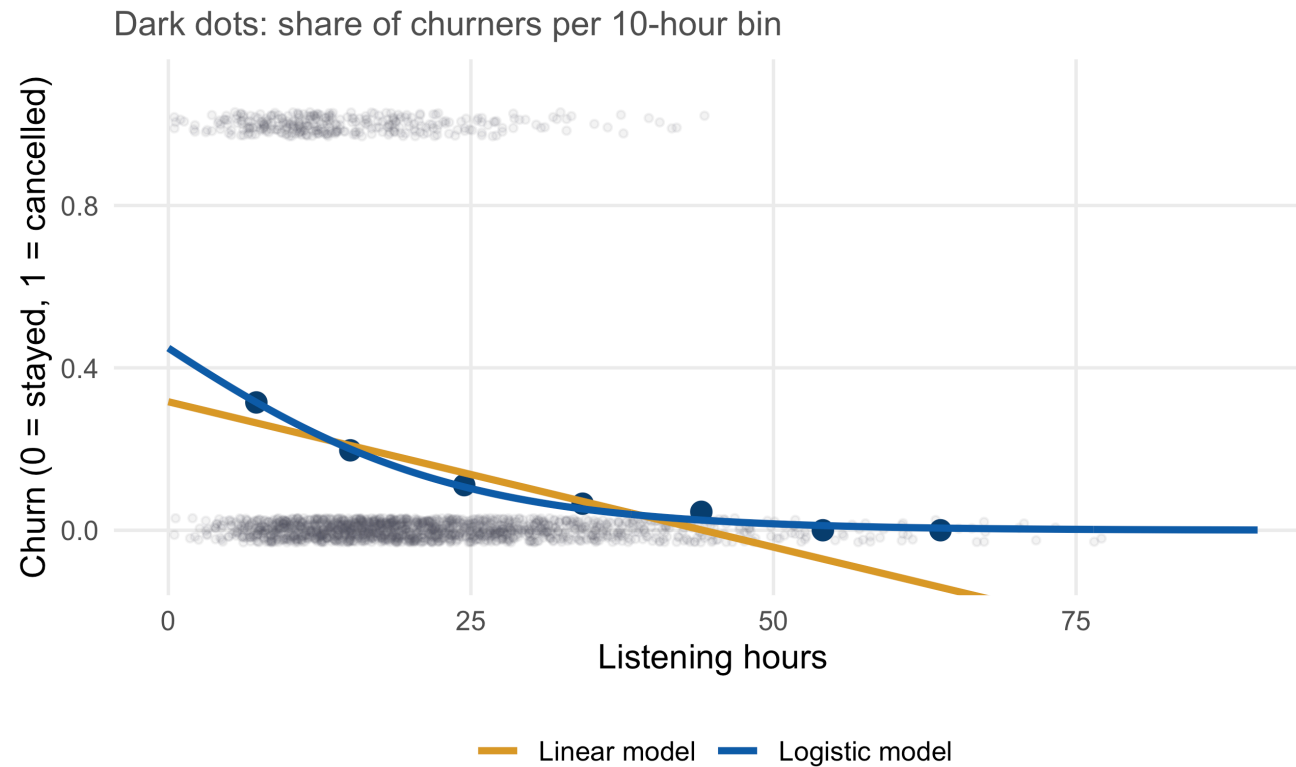
<https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-form>

Screenshot placeholder: the functional-form widget
`img/ch05/widget_loglog.png`

Elasticity < 1 : diminishing returns, the line misses at both ends. $= 1$: proportional. > 1 : increasing returns (rare — network effects).

5.8 Yes/no outcomes: logistic regression

Why not a straight line?



- Outcome: **churn** (cancelled within 3 months), 17% of users
- A line produces probabilities **below 0** for heavy listeners
- The **logistic** curve stays between 0 and 1 — and bends where it matters

```
1 model_churn <- glm(churn ~ hours + premium + tenure + smar
2                     family = binomial, data = smartmix_obs)
```

Reading the coefficients: through probabilities

```
1 model_churn <- glm(churn ~ hours + premium + tenure + smart_mix, family = binomial, data = smartmix_obs)
2 coef(summary(model_churn))
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.27712	0.161358	1.717	8.590e-02
hours	-0.06066	0.008507	-7.130	1.001e-12
premium	-0.53299	0.152782	-3.489	4.856e-04
tenure	-0.02045	0.003752	-5.452	4.983e-08
smart_mix	-0.25635	0.162344	-1.579	1.143e-01

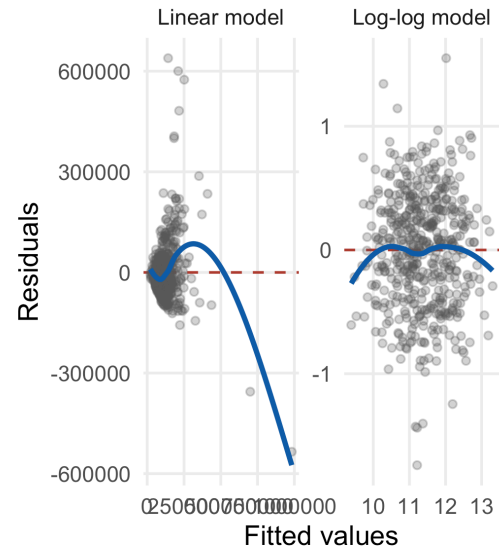
```
1 typical_user <- data.frame(hours = c(10, 20, 30), premium = 0,
2                               tenure = 24, smart_mix = 0)
3 predict(model_churn, newdata = typical_user, type = "response")
```

1	2	3
0.3057	0.1936	0.1157

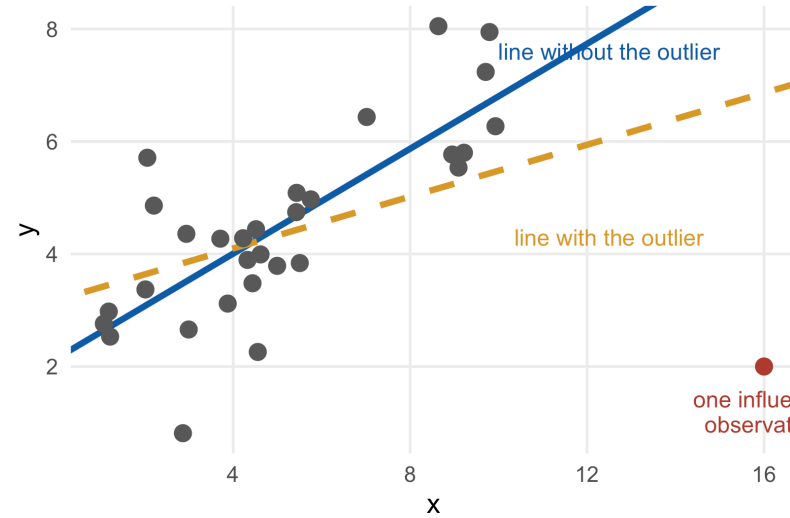
Read **sign** and **significance** directly; read **size** through **predicted probabilities**: 10 vs. 30 hours → churn probability 31% vs. 12%. Differences are effects in **percentage points**. Odds ratios: book box.

5.9 The diagnostics that change decisions

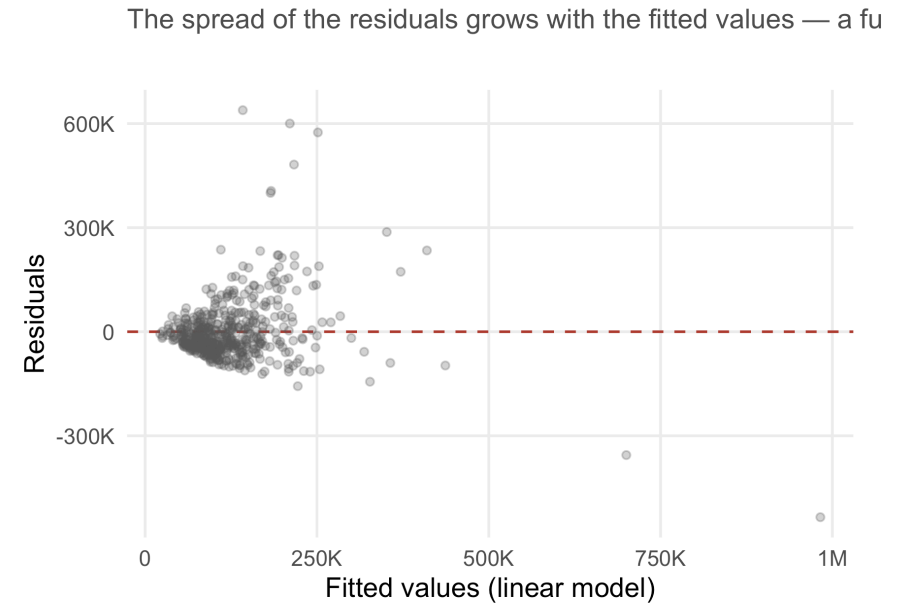
Three checks worth doing every time



1. Is the line the right shape? Curved residuals → change the functional form (log) or add a predictor.



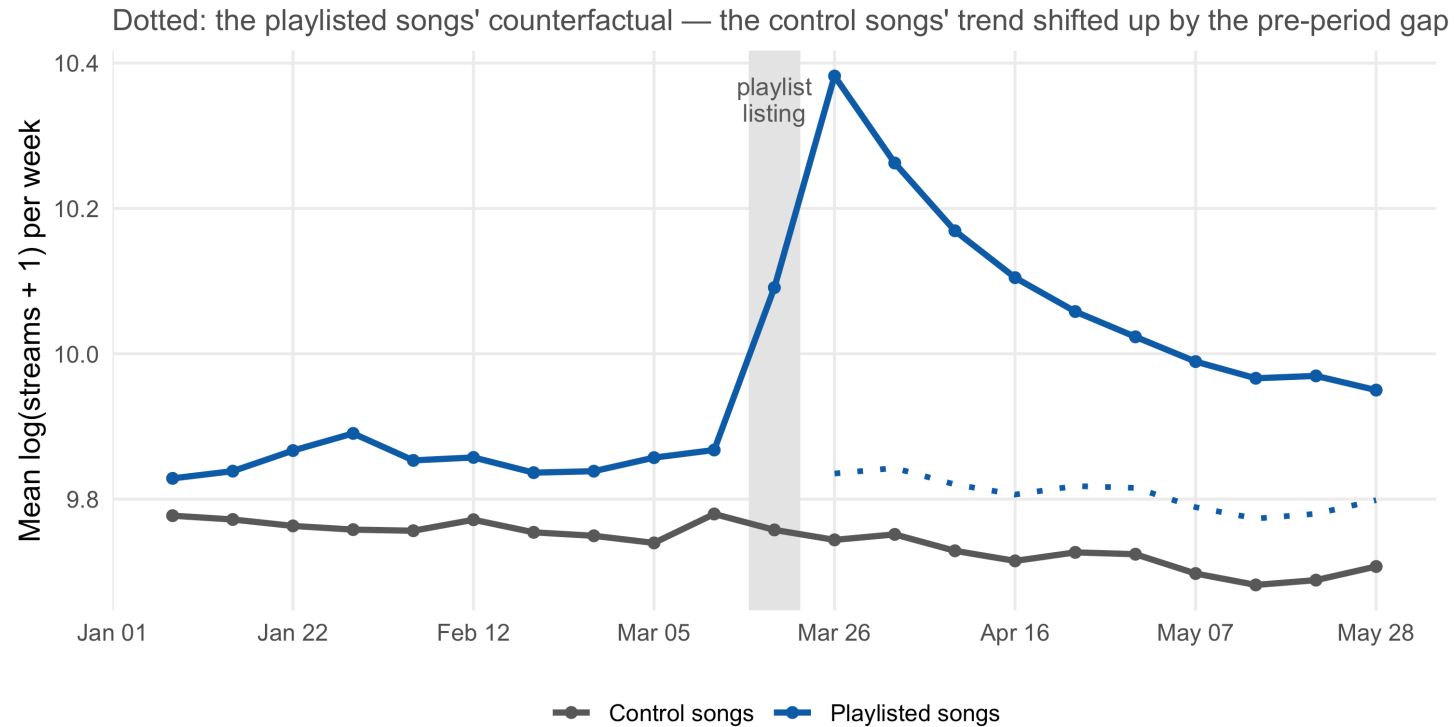
2. Does one observation drive the result? `plot(model, which = 5)`; refit without it — if the conclusion changes, say so.



3. Is the uncertainty right? A funnel → **robust standard errors:**
`coeftest(model, vcov = vcovHC(model))`.

5.10 A quasi-experiment: difference-in-differences

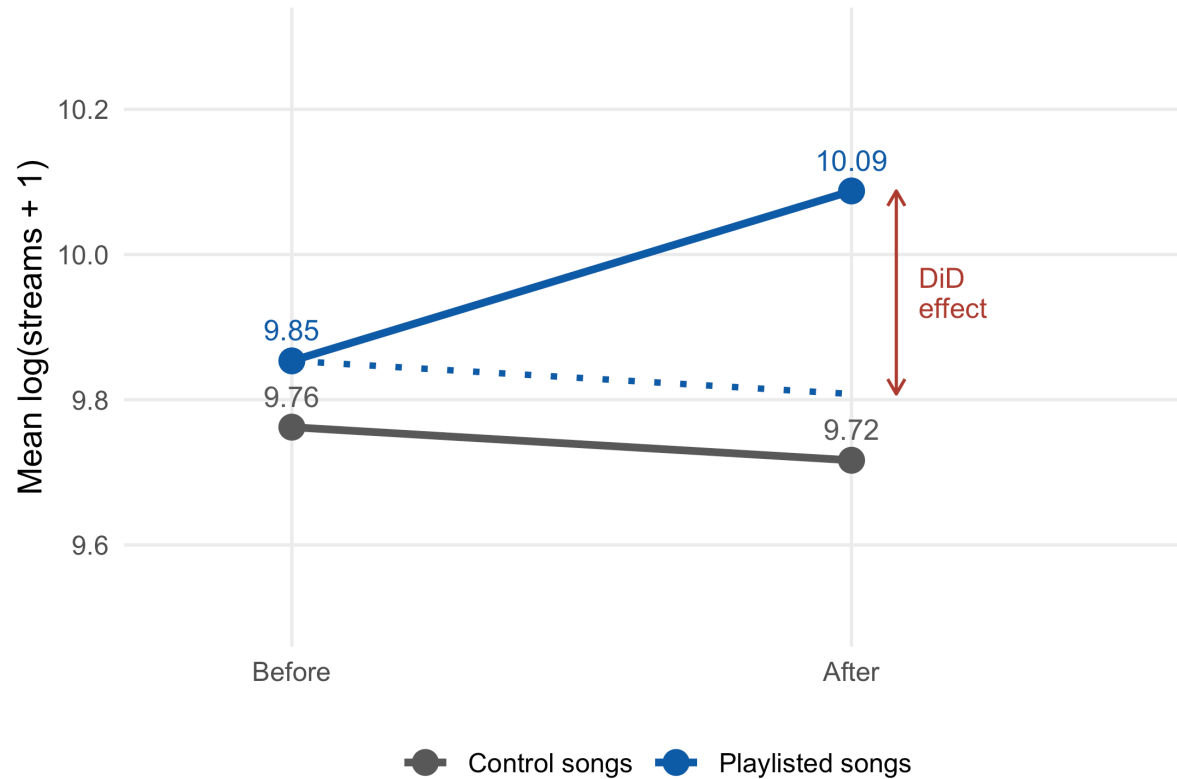
Playlist listings as a natural experiment



- **Real data:** 290 songs × 21 weeks
- March 19, 2018: **53 songs** added to a major playlist; 237 control songs
- Treated songs start **higher** → levels not comparable
- Before the listing: both groups move in **parallel**
- At the listing: the treated group **jumps**

Data: WU-RDS research project (github.com/WU-RDS/did).

Two differences, one estimate



$$\text{DiD} = (\bar{y}_{T,\text{after}} - \bar{y}_{T,\text{before}}) - (\bar{y}_{C,\text{after}} - \bar{y}_{C,\text{before}})$$

$$(10.09 - 9.85) - (9.72 - 9.76) = \mathbf{0.28 \text{ log points}}$$

The control group's change estimates what would have happened to the treated songs **without** the listing — the counterfactual (dotted).

The DiD regression

```
1 did_data <- playlist |>
2   filter(week != as.Date("2018-03-19")) |>      # drop the listing week
3   mutate(log_streams = log(streams + 1))
4 model_did <- lm(log_streams ~ treated * post, data = did_data)
5 coef(summary(model_did))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.76220	0.01788	545.865	0.000000000
treated	0.09128	0.04183	2.182	0.029152094
post	-0.04560	0.02529	-1.803	0.071430068
treated:post	0.27964	0.05916	4.727	0.000002336

Coefficient

Meaning

(Intercept) 9.76

Control, before: baseline

treated 0.09

Treated, before: the **level difference**

post -0.05

Control, after – before: the **common time trend**

treated:post 0.28

The extra change of the treated songs: the **DiD effect** → $\approx +32\%$ streams

The assumption: parallel trends

```
1 placebo_data <- did_data |>
2   filter(post == 0) |>
3   mutate(fake_post = ifelse(week >= as.Date("2018-02-12"), 1, 0))
4   coef(summary(lm(log_streams ~ treated * fake_post,
5                   data = placebo_data)))["treated:fake_post", ]
```

Estimate	Std. Error	t value	Pr(> t)
0.002308	0.085282	0.027059	0.978414

Without the listing, the treated songs would have followed the **same trend** as the controls. **Untestable** — but the pre-period makes it plausible: a fake listing in the pre-period shows **no effect**.

Live demo: course book, \$5.10

<https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-did>

Your turn (5 min)

Add `major_label` as a control variable to the DiD regression:

```
1 coef(summary(lm(log_streams ~ treated * post + major_label, data = did_data)))
```

Does the effect change? Then: name one situation in which the parallel-trends assumption would **fail** for playlist listings.

What observational data can and cannot tell you

Two strategies, one honest report

- **Adjustment** handles confounders you can **measure** — only as good as your proxies; “enough” is untestable
- **Difference-in-differences** handles confounders **constant over time** — needs parallel trends, plausible but never proven
- **Prediction** needs no causal assumptions — a good predictor is valuable even if no coefficient is an effect

Report effect + uncertainty + **assumptions**: “Adjusting for prior listening, tenure, Premium and device, adopters listen $\approx$ 4.5 hours more (95% CI 3.6–5.3), assuming no further differences affect listening — the experiment (3.2) suggests this is not fully met.”

Assignment 3: analyzing observational data

- Your **personal** data (based on your student ID): an opt-in feature, promotion campaigns, and a small panel with a listing
- Tasks: naive vs. adjusted · categorical predictor · elasticity · logistic regression · DiD
- **Deadline:** Group A Sun, Nov 22 · Group B Wed, Nov 25
- **Debrief** at the start of the following session

10 points: report 7 + debrief 3. State your **assumptions** — the best answers say what the data *cannot* show.

Learning check (Part II)

1. Log-log coefficient of ad spend = 0.15. Interpret. The level model says 0.8 sales per euro — why might the two disagree about doubling the budget?
2. Log-level model, dummy “promotion week” = 0.35. Percentage difference in sales — approximate and exact?
3. Residual plot shows a funnel. Consequence for coefficients, for p -values — and the simplest fix?
4. A DiD of a price promotion in treated stores: which assumption, and how can the pre-period help?

Next session

Module 6 · Mechanisms & heterogeneity — *Why, and for whom, does it work?*

- Interaction effects: when the effect differs between groups (moderation)
- Mediation: through which mechanism an effect works
- In experiments and observational data — and in your group project